

Luca Walz

Backend and Cloud Engineer

GitHub github.com/lucawalz · LinkedIn linkedin.com/in/luca-walz · lucawalz.dev

Backend and cloud engineer who likes following a system the whole way down, from the API surface to the infrastructure underneath. Most of my work sits around Kubernetes, NixOS and GitOps, with a fair bit of Go, plus LLM agents and full-stack IoT on the side.

EXPERIENCE

IBM · Böblingen, Hybrid 10 mos
Working Student · Part-time Sep 2025 - Dec 2025
Software Development Intern · Full-time Mar 2025 - Aug 2025

Built API test suites and backend testing workflows for IBM Knowledge Catalog, validating enterprise-scale microservices alongside the dev and QA teams.

Java · Jenkins · Microservices · API testing

EDUCATION

Hochschule der Medien, Stuttgart · BSc Media Informatics 2023 - Present
Bachelor thesis on autonomous cluster remediation with LLM agents.

Freie Waldorfschule, Böblingen · Abitur 2020 - 2022

Burnaby Mountain Secondary School, Vancouver · Exchange year, Canada 2019 - 2020

SELECTED PROJECTS

vigil Python, Go
My bachelor thesis: a multi-agent system that diagnoses and repairs faults in a Kubernetes cluster on its own. The LLM agents act through typed MCP tools and are reversible by construction, with Flux GitOps and NixOS generations giving every repair a clean way back.

bedrock Nix
A bare-metal Kubernetes homelab that lives entirely in Git. Three NixOS nodes run K3s, and Flux reconciles the cluster from the repository instead of by hand.

horizon Go
The active half of the homelab: a Go CLI that rents reserved Hetzner Cloud nodes on demand, folds them into the cluster over Tailscale, moves a workload onto them, and tears them back down when they are no longer needed.

sentio-systems · Project Lead and Head Developer Java, TypeScript, Python
A wildlife and weather monitoring platform. Raspberry Pi field devices feed sensor and camera data to a Spring Boot backend, where AI classifies the animals down to the bird species, with a React dashboard and an agentic chatbot to explore it all.

SKILLS

Languages	Go, Python, Java, TypeScript
Runtimes and frameworks	Spring, React, Nix
Infrastructure and data	Linux, Kubernetes, Docker, Terraform, Postgres, Redis, Prometheus, Grafana, Raspberry Pi

VOLUNTEERING

Student Mentor · Hochschule der Medien, Stuttgart Mar 2024 - Aug 2024
Supporting first-semester students in finding their footing on campus and in their studies.

Downhill Trainer Since 2014

Scout Since 2008
Long-term member, organising outdoor excursions with larger groups.

LANGUAGES

German (Native) · English (C2) · Russian (A1)